Vendor Performance Analysis

An End-to-End Business Intelligence Project

Viranchi Ravindra More

March 1, 2026

Contents

1 Business Problem **2**

2 Exploratory Data Analysis (EDA) **2**
2.1 Summary Statistics 2
2.2 Key Observations 3
2.3 Data Cleaning Criteria 4

3 Correlation Analysis **4**

4 Research Findings **6**
4.1 Brands for Promotional or Pricing Adjustments 6
4.2 Vendor Contribution Analysis 7
4.3 Impact of Bulk Purchasing 8
4.4 Low Inventory Turnover Vendors 8
4.5 Profit Margin Comparison 9

5 Statistical Validation **10**

6 Final Recommendations **10**

7 Conclusion **11**

1 Business Problem

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies must prevent losses caused by inefficient pricing, poor inventory turnover, or excessive vendor dependency.

The objectives of this analysis are:

- Identify underperforming brands requiring promotional or pricing adjustments.
- Determine top vendors contributing to total sales and gross profit.
- Analyze the impact of bulk purchasing on unit cost reductions.
- Assess inventory turnover efficiency to reduce holding costs.
- Compare profitability between high-performing and low-performing vendors.

2 Exploratory Data Analysis (EDA)

2.1 Summary Statistics

Initial exploration revealed significant variance across pricing, sales, and profitability metrics.

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

Figure 1: Summary Statistics

2.2 Key Observations

Negative and Zero Values:

The dataset contains several financially significant negative and zero-value observations:

- **Gross Profit Minimum (-52,002.78):** Indicates certain transactions were executed at a loss, possibly due to heavy discounts, high procurement costs, or pricing inefficiencies.
- **Profit Margin Minimum ($-\infty$):** Occurred in cases where revenue was zero, leading to undefined margins. This highlights instances of purchased inventory that was never sold.
- **Zero Sales Quantity:** Some products show purchases without corresponding sales, suggesting slow-moving or obsolete inventory that increases holding costs and reduces capital efficiency.

Outliers and Variability:

Significant variation was observed across pricing, logistics, and turnover metrics:

- **Purchase and Actual Prices:** Maximum values (5,681.81 and 7,499.99) are substantially higher than average levels, indicating premium or specialty product segments.
- **Freight Cost (0.09 to 257,032.07):** Large dispersion suggests logistical inconsistencies, bulk shipment effects, or vendor distance variability.
- **Stock Turnover (0 to 274.5):** Wide range indicates uneven inventory movement, with some products selling rapidly while others remain stagnant.

These findings emphasize the need for improved pricing strategies, inventory optimization, and vendor performance monitoring.

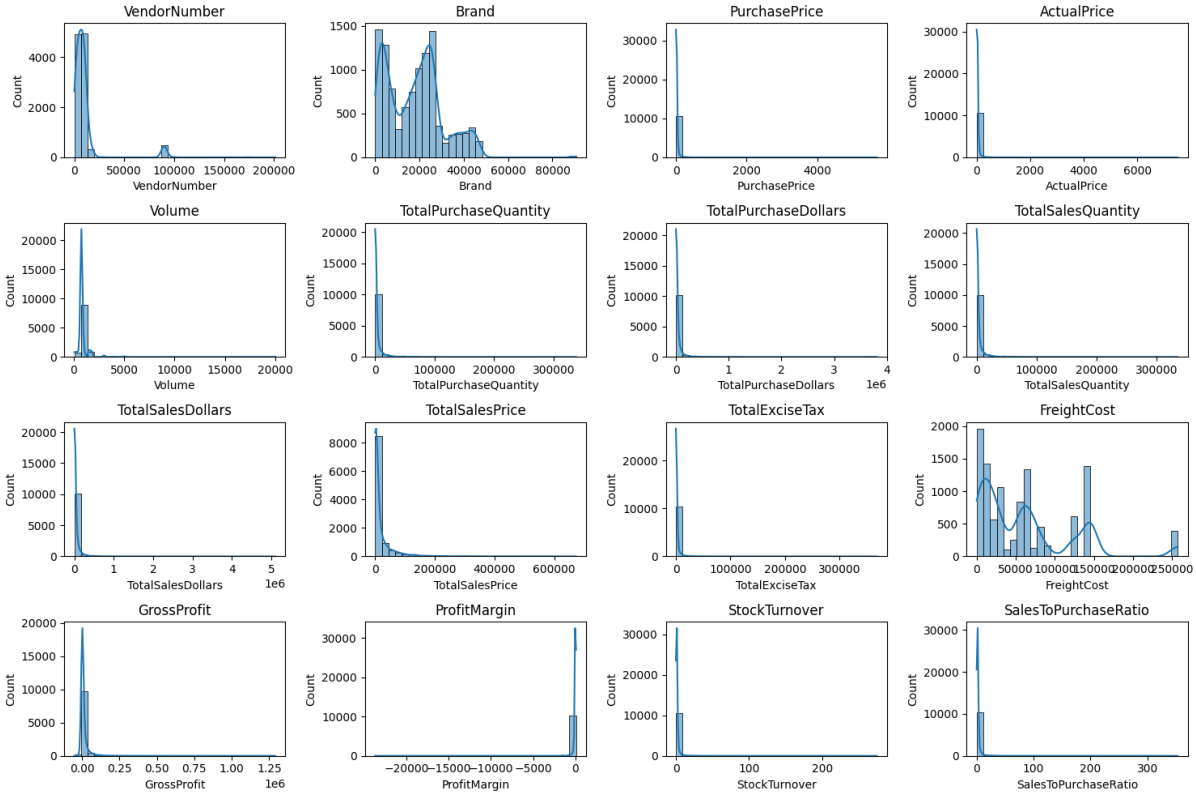

Figure 2: Distribution Plots for Numerical Columns

2.3 Data Cleaning Criteria

To improve reliability:

- Removed records with Gross Profit ≤ 0
- Removed records with Profit Margin ≤ 0
- Removed products with Total Sales Quantity $= 0$

3 Correlation Analysis

The correlation analysis provides insight into relationships between pricing, sales, and profitability metrics:

- **Purchase Price vs Sales Dollars (-0.012):** The near-zero correlation indicates that variations in purchase price have minimal impact on total sales revenue. This suggests demand is relatively price-insensitive or influenced by other factors such as brand strength or market positioning.
- **Purchase Quantity vs Sales Quantity (0.999):** The extremely strong positive correlation confirms efficient inventory movement. Products purchased in higher

volumes are generally sold in comparable quantities, indicating effective stock planning and demand alignment.

- **Profit Margin vs Sales Price (-0.179):** The weak negative relationship suggests that higher sales prices do not necessarily translate into higher margins, possibly due to competitive pricing pressures or increased associated costs.

- **Stock Turnover vs Profitability:** The weak negative association indicates that faster inventory movement does not automatically guarantee higher profitability. This highlights the importance of balancing sales velocity with margin optimization.

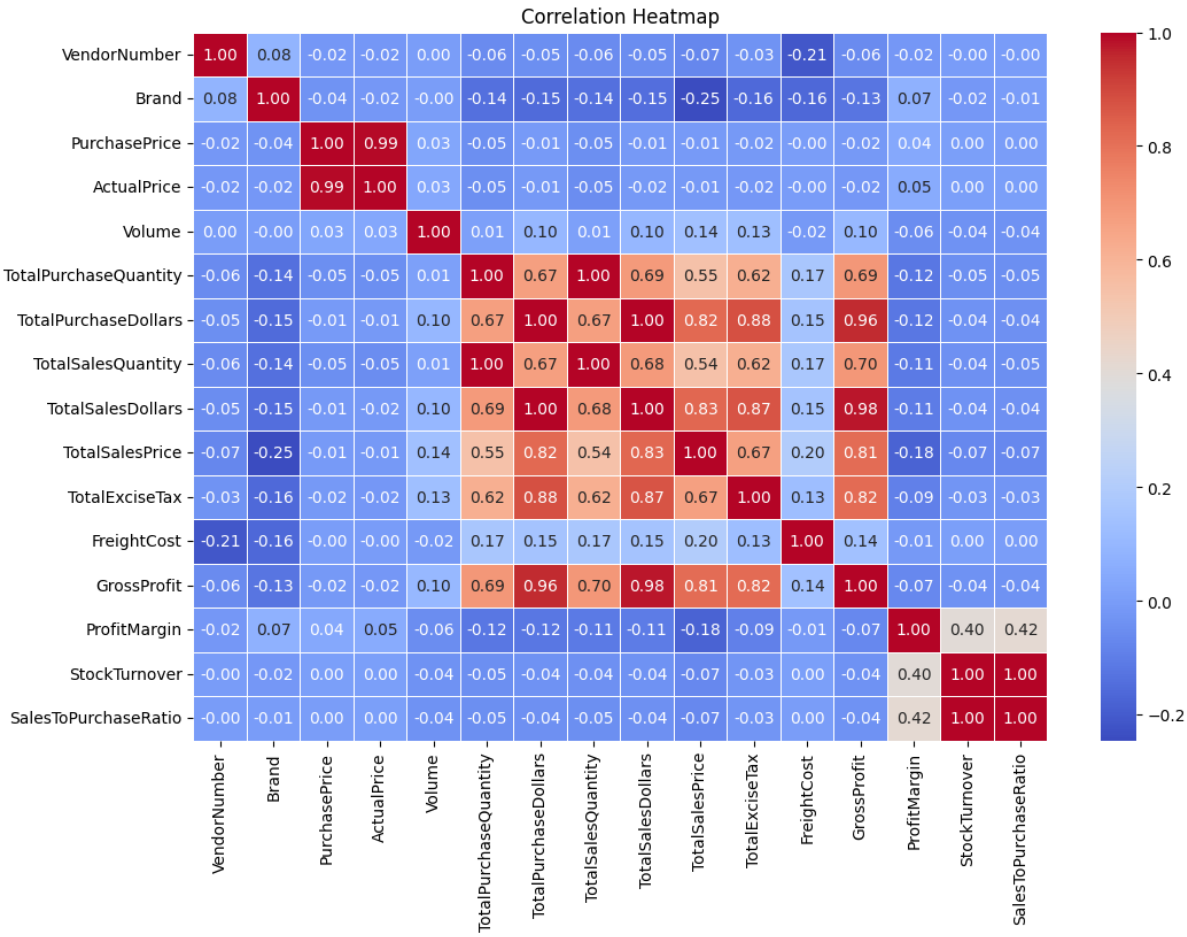

Figure 3: Correlation Heatmap of Key Variables

4 Research Findings

4.1 Brands for Promotional or Pricing Adjustments

The analysis identified 198 brands characterized by relatively low sales volumes but high profit margins. This combination suggests that while these brands are profitable on a per-unit basis, they may not be reaching sufficient market demand.

These brands present an opportunity for strategic growth through:

- **Targeted Promotions:** Increasing brand visibility through marketing campaigns or seasonal promotions to drive higher sales volume.
- **Price Optimization:** Slight price adjustments to stimulate demand while maintaining healthy margins.
- **Improved Distribution:** Expanding channel presence or optimizing placement to improve accessibility and customer reach.

By improving sales velocity without significantly compromising margins, these brands can contribute more effectively to overall revenue growth.

```
Brands with Low Sales but High Profit Margins:
```

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

198 rows × 3 columns

Figure 4: Brands with Low Sales but High Profit Margins

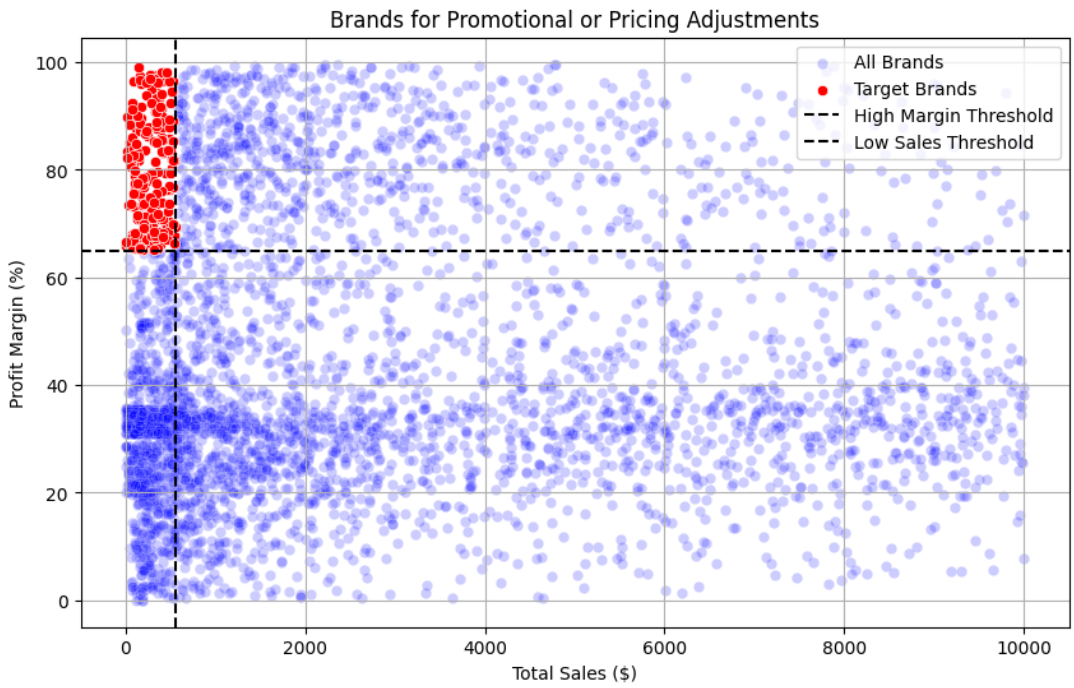

Figure 5: Brands for Promotional or Pricing Adjustments

4.2 Vendor Contribution Analysis

Top 10 vendors contribute 65.69% of total purchases. This concentration risk suggests potential supply chain dependency.

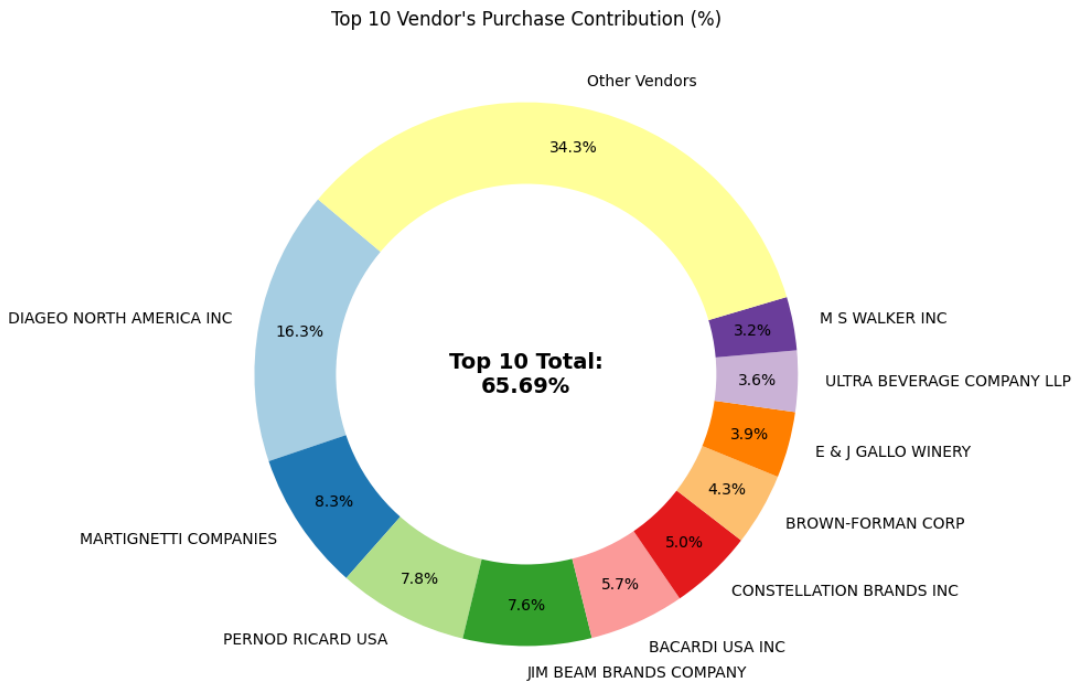

Figure 6: Top 10 Vendors Purchase Contribution

4.3 Impact of Bulk Purchasing

Large order sizes reduce unit purchase cost by 72%.

Order Size	Avg Unit Purchase Price
Small	39.06
Medium	15.48
Large	10.78

Table 1: Bulk Purchase Cost Advantage

4.4 Low Inventory Turnover Vendors

Total Unsold Inventory Capital: $2.71M.

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

Figure 7: Unsold Inventory Capital

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

	VendorName	StockTurnover
0	ALISA CARR BEVERAGES	0.615385
36	HIGHLAND WINE MERCHANTS LLC	0.708333
60	PARK STREET IMPORTS LLC	0.751306
19	Circa Wines	0.755676
26	Dunn Wine Brokers	0.766022
15	CENTEUR IMPORTS LLC	0.773953
78	SMOKY QUARTZ DISTILLERY LLC	0.783835
90	TAMWORTH DISTILLING	0.797078
91	THE IMPORTED GRAPE LLC	0.807569
101	WALPOLE MTN VIEW WINERY	0.820548

Figure 8: Vendors with Low Inventory Turnover

4.5 Profit Margin Comparison

A comparative analysis was conducted between top-performing vendors (based on sales volume) and low-performing vendors to evaluate profitability differences.

Top Vendors:

- Mean Profit Margin: 31.17%
- 95% Confidence Interval: (30.74%, 31.61%)

Low Vendors:

- Mean Profit Margin: 41.55%
- 95% Confidence Interval: (40.48%, 42.62%)

The results indicate that low-performing vendors maintain significantly higher profit margins compared to top-selling vendors. However, their limited sales volume restricts overall revenue contribution. In contrast, top vendors operate with lower margins but generate substantial revenue through higher sales volumes.

This contrast highlights differing business models: high-volume, lower-margin strategies versus low-volume, higher-margin positioning.

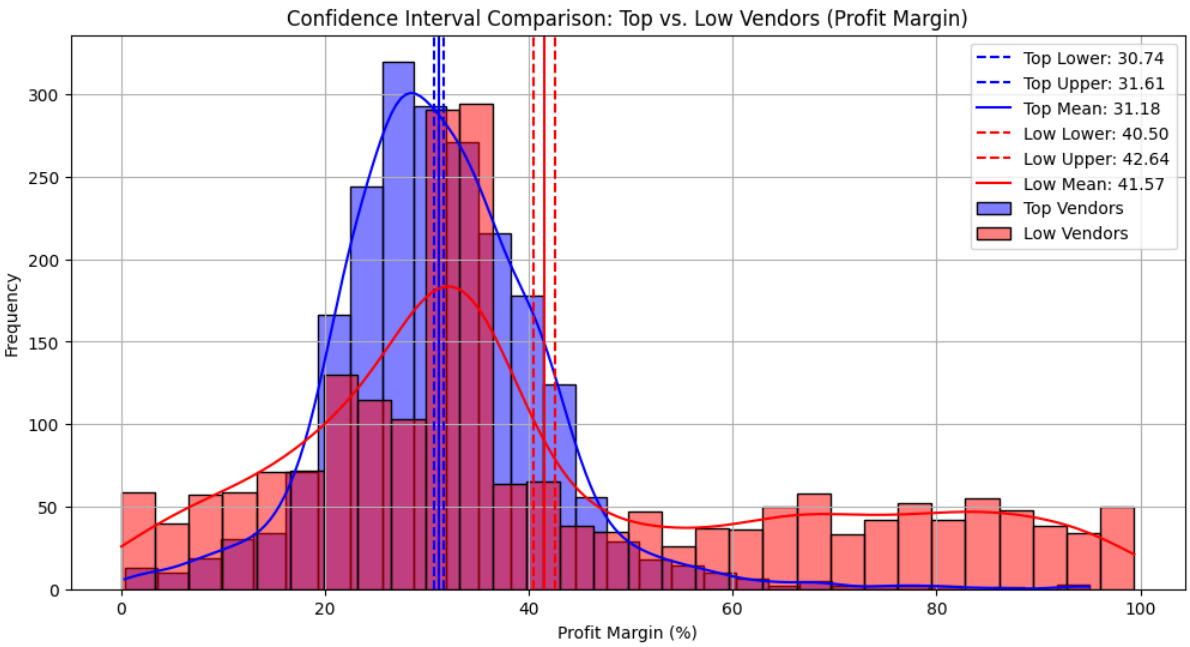

Figure 9: Confidence Interval Comparison for Profit Margin: Top vs Low Vendors

5 Statistical Validation

Null Hypothesis (H_0): No significant difference in profit margins.

Alternative Hypothesis (H_1): Significant difference exists.

Result: Null hypothesis rejected.

Conclusion: Vendor groups operate under distinct profitability models.

6 Final Recommendations

Based on the analytical findings, the following strategic actions are recommended:

- **Re-evaluate Pricing Strategies:** Adjust pricing for low-sales, high-margin brands to stimulate demand while preserving profitability.
- **Diversify Vendor Portfolio:** Reduce over-reliance on a small group of vendors to mitigate supply chain and operational risks.
- **Leverage Bulk Purchasing:** Capitalize on volume-based cost reductions to improve unit economics and strengthen competitive positioning.
- **Optimize Inventory Management:** Reduce slow-moving inventory through demand forecasting improvements, clearance initiatives, and purchase quantity adjustments.

- **Strengthen Market Reach:** Enhance marketing and distribution strategies for underperforming vendors to increase sales volume without significantly compromising margins.

7 Conclusion

This analysis demonstrates how data-driven decision-making can significantly improve vendor performance, pricing efficiency, and inventory management.

By balancing sales volume with profit margins, optimizing procurement strategies, and addressing inventory inefficiencies, organizations can unlock working capital, mitigate operational risks, and drive sustainable profitability.

The integration of statistical analysis and business intelligence tools provides a scalable framework for continuous performance monitoring and strategic optimization.